LABORATORY HANDOUT : Generate a Lexical Analyzer Using Lex (or Flex)

1. Shell commands to generate a lexical analyser from a lex script

Write the regular expressions for tokens of interest in the format of lex specifications, which is known as the lex script. Perform the following steps to generate the scanner (also called as lexical analyser).

Step 1 : Run the lex script through lex as shown below.

```
$ lex exp1.l
```

In case there are no errors in the script, lex generates a C program in the file named as “lex.yy.c”. You can read the contents of this generated file now or later by running the ls command.

```
$ ls −l
```

Step 2 : Compile the lex.yy.c program using gcc.

```
$ gcc lex.yy.c -o scan1
```

On successful compilation the executable code is generated by gcc, named as “scan1”, using option “-o” and “scan1” is our executable scanner.

Step 3: For ease of experimentation, you may use relevant features of the shell, such as i/o redirection for using scan1 over the different input files.

```
$ ./scan1 < input1
```

in which case the output of the scanner is displayed on screen. In case you want to save the output of the scanner in a file, use $ ./scan1 < input1 > output1

You can easily save the inputs and outputs together in your directory so that your work done during the lab session is recorded for future.

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